

What happens when coupling switches on?

exp085 · 17 August 2026 · Draft

Abstract

Two slightly detuned PING circuits begin from the same mature uncoupled trajectory before reciprocal excitation switches on. Across 11 coupling strengths, the unwrapped relative phase changes from a persistent drift of 30.1 rad/s to -0.01 rad/s. The experiment uses runtime-state branching to isolate coupling onset from network activation and shows the transition directly without imposing a binary synchronization threshold.

Methods

1. **Equilibrate two detuned circuits.** Independent 100 Hz/channel Poisson streams drive two 80 E / 20 I PING components. Circuit A uses 4 ms inhibitory decay and circuit B uses 5 ms. Both run uncoupled for 1000 ms.
2. **Branch one mature state.** The executor captures voltages, refractory counters, conductances, population histories, and input-delay histories at coupling onset. That identical state branches across 11 reciprocal coupling strengths. Private future inputs remain paired across K and independent between circuits.
3. **Measure relative phase.** E-population rates are smoothed with a 5 ms Gaussian kernel and filtered over 20–60 Hz. Hilbert phases define the unwrapped A-minus-B relative phase, expressed as its change from coupling onset. The final 100 ms are excluded to prevent the finite-record filter boundary from appearing as a phase jump. No locking threshold or latency is inferred.
4. **Measure supporting responses.** The registered `exp085-dominant-rhythm-v1` policy reports each circuit’s dominant rhythm from the final 1.5 s. Frequency difference, descriptive phase-locking value, and population firing rates support the phase trajectories. Figure 3 measures every trial at one illustrative K by its circular phase error from its own mean terminal offset, defined over the final 500 ms and smoothed over 50 ms. The displayed K is the condition containing the stable trial with the largest clean onset-to-settled reduction; all samples at that K are then shown. This is descriptive, not a registered latency estimator.

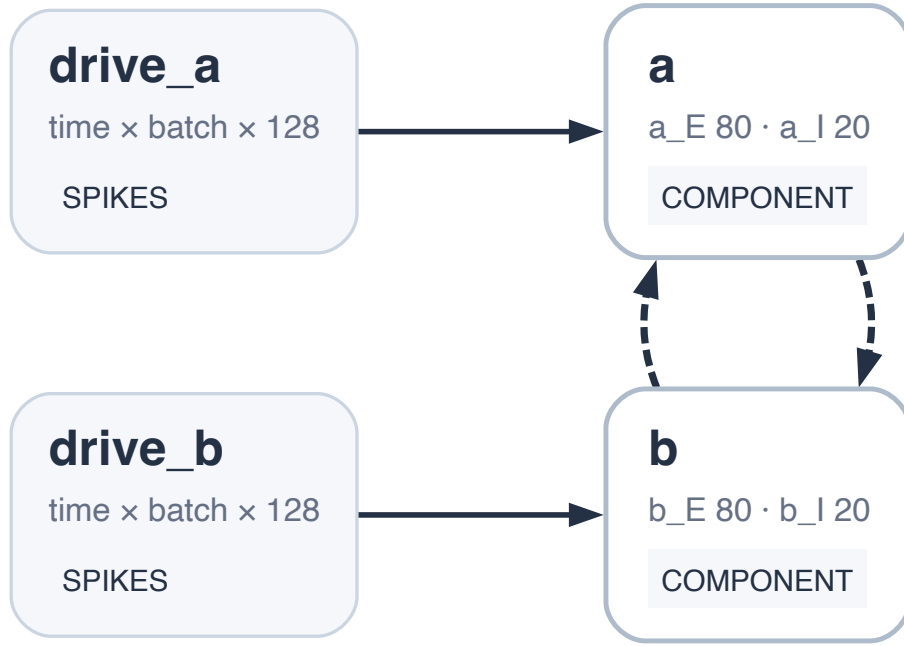


Figure 1: Two-circuit coupling-onset graph. Independent spike inputs drive circuits A and B. Four reciprocal AMPA projections share coupling strength K ; their weights are zero during equilibration and vary only after the common runtime-state branch. Local PING topology and every non-coupling parameter remain fixed.

Results

The coupling scan shows frequency convergence and rising phase concentration without reducing either to a binary label. At $K = 0$ the circuits repeatedly lap one another; the strongest condition ends with a median phase-locking value of 0.99.

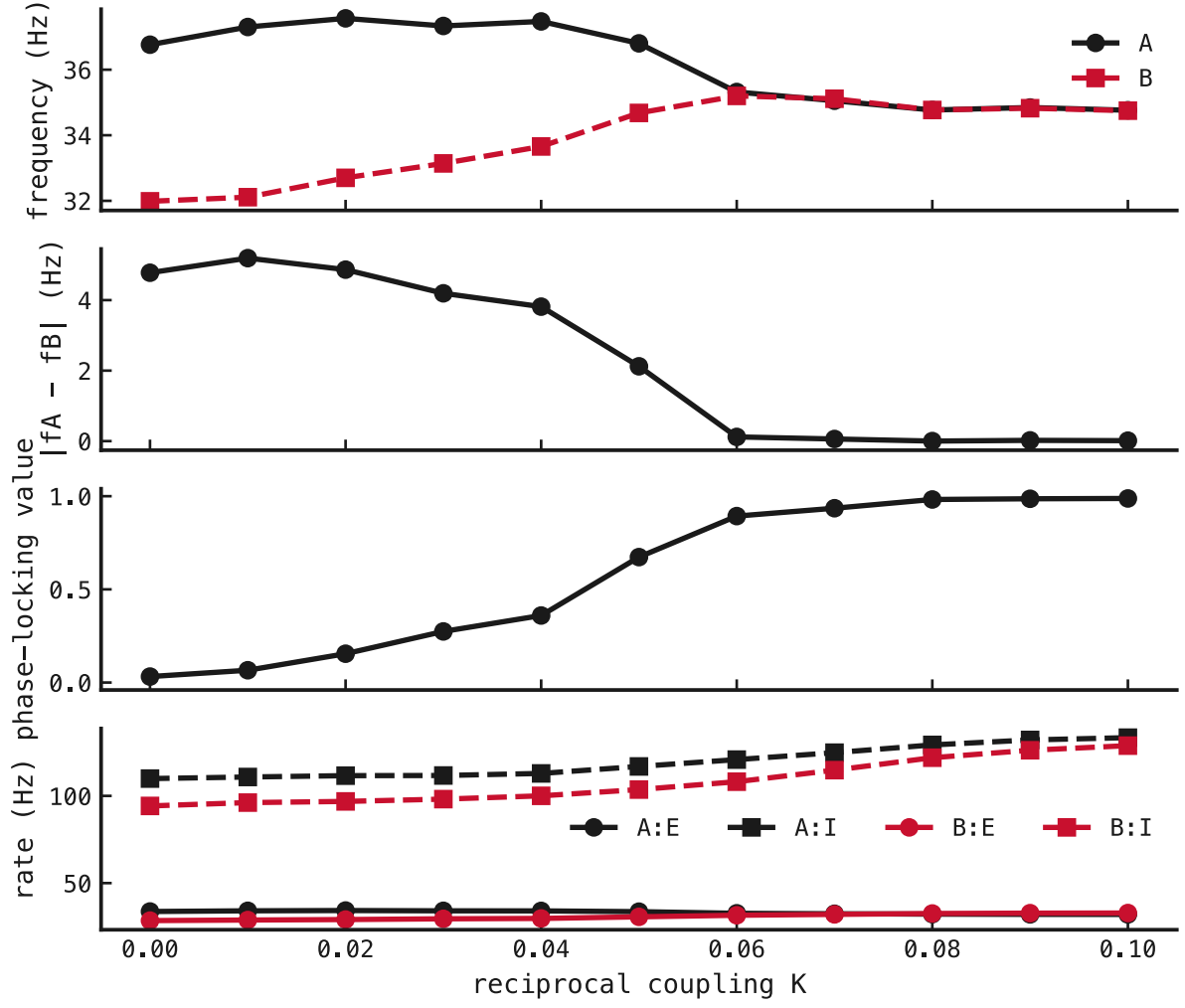


Figure 2: Supporting coupling response. First: median dominant E-population frequencies of circuits A and B in hertz. Second: their absolute difference in hertz. Third: median descriptive phase-locking value across 5 paired trials. Fourth: mean E and I firing rates in hertz for both circuits. K is the shared weight of the four reciprocal AMPA projections.

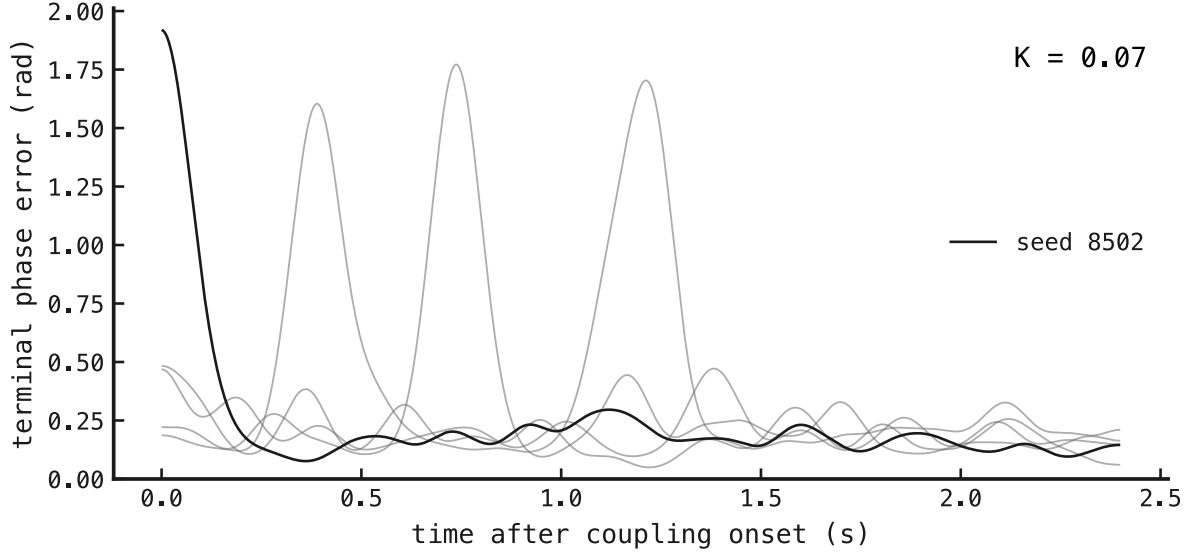


Figure 3: All 5 paired trials at the selected $K = 0.07$. Each trace is that trial’s circular error from its own mean phase offset during the final 500 ms, with 50 ms smoothing; lower values indicate closer approach to the terminal relationship. The rule-selected seed 8502 is black and the other seeds are grey. Retrospective terminal references make these illustrations of convergence, not synchronization-latency estimates.

Under the same illustrative clean-convergence rule, the passing counts are $K = 0$: 0/5; $K = 0.01$: 0/5; $K = 0.02$: 0/5; $K = 0.03$: 0/5; $K = 0.04$: 0/5; $K = 0.05$: 0/5; $K = 0.06$: 0/5; $K = 0.07$: 2/5; $K = 0.08$: 5/5; $K = 0.09$: 4/5; $K = 0.1$: 5/5. A trial passes only when its terminal-error 95th percentile is at most 0.25 rad and it has no post-settling excursion above 0.5 rad. This post-hoc descriptive rule is not a formal synchronization boundary.